

Multiple Choice

1. **Answer:** D

Options: A) square; B) hexagon; C) cube; D) tetrahedron

Note: Correct option: D - tetrahedron

2. **Answer:** B

Options: A) 1.6 ns; B) 2.5 ns; C) 4.2 ns; D) 6.9 ns

Note: Correct option: B - 2.5 ns

3. **Answer:** B

Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic

Note: Correct option: B - Constant volume

4. **Answer:** C

Options: A) 1.2×10^9 m/s; B) 3.0×10^8 m/s; C) 1.5×10^8 m/s; D) 1.0×10^8 m/s

Note: Correct option: C - 1.5×10^8 m/s

5. **Answer:** B

Options: A) 0.50 c; B) 0.60 c; C) 0.70 c; D) 0.80 c

Note: Correct option: B - 0.60 c

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' 1.5×10^8 m/s'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'an average of 10 times, with an rms deviation of about 3'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The entropy of the system and its environment remains unchanged.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 393 Hz. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 612 Hz. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key